

DATASCI 503: Statistical Machine Learning

Winter 2026

Instructor: Jeffrey Regier

Class time: Tuesdays & Thursdays, 10 am – 11:30 am

Class location: School of Education Building (SEB), Room # 1202

Course Canvas website: <https://umich.instructure.com/courses/805526>

Instructor email: regier@umich.edu (Please see “How to get help” below before emailing.)

Office hours: See the “calendar” tab in Canvas.

GSI information & lab sections

Lab sections are held on Fridays beginning January 16, 2025.

SECTION 101

GSI: Daniel Zou

Lab time: 11:30 am – 1 pm

Lab place: 260 Weiser Hall

GSI email: zoudj@umich.edu

SECTION 102

GSI: Paolo Borello

Lab time: 11:30 am – 1 pm

Lab place: 296 Weiser Hall

GSI email: borello@umich.edu

GSI office hours are posted on Canvas. Please read the section below titled “How to get help” before emailing your GSI.

Course description

This course covers modern statistical learning methods, balancing theory and practice. Topics include foundational concepts (bias-variance trade-off, curse of dimensionality, cross-validation), supervised methods for classification and regression (nearest neighbors, discriminant analysis, logistic regression, splines, generalized additive models, random forests, boosting, support vector machines, neural networks, CNNs), and unsupervised methods (agglomerative clustering, k-means, variational autoencoders). Students will learn what these methods do, when to use them, and why they work.

Prerequisites

This course assumes familiarity with regression analysis (STATS 500 or equivalent), linear algebra (MATH 214 level), and theoretical statistics (STATS 426 level).

Prior experience with Python is not required. However, students without much Python experience will need to catch up during the first few weeks. For these students, a good reference is *Python for Everybody* by Charles Severance.

Textbooks

G. James, D. Witten, T. Hastie, R. Tibshirani, and J. Taylor (2023). *An Introduction to Statistical Learning with Applications in Python (ISLP)*. Springer.

T. Hastie, R. Tibshirani, and J. Friedman (2009). *The Elements of Statistical Learning: Data*

Mining, Inference, and Prediction, Second edition. Springer.

S. J. D. Prince (2024). *Understanding Deep Learning (UDL)*. MIT Press.

All three books are available for free download through the links above.

For most of the course, we will proceed linearly through ISLP chapters 1–10 and 12. Students should read the relevant chapter both before and after each lecture.

Tentative Course Schedule

Week	Dates	Topics	Reading
1	Jan 8	Course overview; bias-variance tradeoff, KNN	ISLP Ch 2
2	Jan 13, 15	Bias-variance tradeoff, KNN (cont.)	ISLP Ch 2
3	Jan 20, 22	Classification: logistic regression	ISLP Ch 4
4	Jan 27, 29	Classification: evaluation metrics, LDA, QDA	ISLP Ch 4
5	Feb 3, 5	Naive Bayes; cross-validation	ISLP Ch 4–5
6	Feb 10, 12	Bootstrap; splines, loess, GAMs	ISLP Ch 5, 7
7	Feb 17, 19	GAMs (cont.); tree-based methods	ISLP Ch 7–8
8	Feb 24, 26	Tree-based methods (cont.); support vector machines	ISLP Ch 8–9
9	Mar 2–6	Winter Break — No Class	
10	Mar 10, 12	Midterm review; midterm exam	
11	Mar 17, 19	PCA; clustering	ISLP Ch 12
12	Mar 24, 26	Neural networks; PyTorch	ISLP Ch 10
13	Mar 31, Apr 2	PyTorch (cont.); feedforward networks, regularization	UDL Ch 1, 3–4
14	Apr 7, 9	Deep networks, double descent; transformers	UDL Ch 8–9, 12
15	Apr 14, 16	Transformers (cont.); final exam review	UDL Ch 12

Grading

Course grades are based on homework (20%), group work (15%), in-class participation through Top Hat (5%), the midterm exam (20%), and the final exam (40%). The distribution of final grades will be similar to that of previous offerings of the course: 45% A/A+, 70% A-/A/A+, and 98% B- and above.

Additionally, bonus points worth up to half of a letter grade will be awarded to students who regularly participate verbally in class.

Homework

There will be weekly homework assignments consisting of data analysis, to be done in Python, and conceptual and/or derivation questions. The lowest homework score will be dropped; therefore, late homework is not accepted. Homework will be completed and submitted electronically through Canvas.

Group work

During weekly lab sections, students will collaborate in two-person teams to complete data analysis in Jupyter notebooks. Initially, teams will be randomly assigned by the GSIs, but students will later be invited to choose their own teammates. GSIs will assist individual teams that encounter

difficulties. Group-work assignments will generally be due a few days after the lab, but many group-work assignments may be completed entirely during the lab section.

The purpose of this format is to teach students about teamwork and collaborative data science practices, which are increasingly important in industry. Working in a group setting can enhance motivation for some students, whereas others benefit from serving as mentors.

Attendance at the lab sections is mandatory; group work not begun in the lab section will not be graded. To accommodate short-term illness, occasional scheduling conflicts, and problematic team assignments, the lowest three group-work scores will be dropped. Missing additional labs without penalty requires the approval of the Dean's Office or the Office of Services for Students with Disabilities (SSD).

In-class participation through Top Hat

During each lecture, I will pose several questions to the class, typically multiple-choice, related to the material covered. Students respond to the questions through the [Top Hat Tests - LSA](#) tab in Canvas and see how their answers compare to their classmates'. These questions are graded on completeness, not correctness. The goal is to help students keep up with the material and to alert them to gaps in their understanding. The questions also help me gauge what is and isn't being understood.

Attending lectures is effectively mandatory; the Top Hat questions can only be answered by students present in person. To accommodate illness, scheduling conflicts, and other legitimate absences, scores for up to five missed lectures will be automatically dropped. Missing additional lectures without penalty requires the approval of the Dean's Office or SSD.

Exams

The midterm exam will be administered in class on March 12, 2026. The final exam will be administered at the university-designated time: Wednesday, April 28, 2026, from 1:30 pm to 3:30 pm.

Both exams are closed-book and do not involve a computer. You are allowed to bring one two-sided page of handwritten notes and a calculator. Neither exam tests knowledge of Python functions, though either may require understanding Python output. A sample midterm and final exam will be posted on Canvas in advance. The final exam is cumulative.

If you have SSD-approved accommodations for exams, please submit your documentation at the beginning of the term. There are no alternate exam dates unless necessary for SSD accommodations.

AI use

AI is a transformative technology, and AI use will likely be a significant part of many of your future jobs. Your education should prepare you for this. In this course, I encourage you to use AI to look up Python syntax, debug errors when you are stuck, improve your grammar, and ask clarifying questions about course content.

However, it is easy to slip from beneficial use to overuse. Our assignments are brain exercises; they only promote growth if you complete them. Even encouraged uses, such as debugging, can become counterproductive if used too readily. A good rule of thumb: if you find yourself prompting AI more than a few times per assignment, or providing any assignment question verbatim, pause and

reconsider. Exams will be without AI assistance, and struggling with assignment problems is the best preparation.

Academic integrity

Academic misconduct includes copying code from the web or from your fellow students, providing code to your fellow students, looking up solutions online, submitting AI-generated solutions, submitting assignments from other classes or previous iterations of this course, and hiring others to complete your work.

From the LSA Community Standards of Academic Integrity:

Academic dishonesty may be understood as any action or attempted action that may result in creating an unfair academic advantage for oneself or an unfair academic advantage or disadvantage for any other member or members of the academic community. Conduct, without regard to motive, that violates the academic integrity and ethical standards of the College community cannot be tolerated.

See <https://lsa.umich.edu/lsa/academics/academic-integrity.html> for more information.

You are welcome to discuss homework with your classmates, but the work you submit must be yours and yours alone. There is one exception to the above policy: you may submit Python code supplied by either GSI during a lab section without attribution.

If you use external sources, you must cite and credit them.

Academic misconduct will be met with disciplinary action. Students who engage in academic misconduct will typically receive a failing grade in this course. Additional sanctions may be imposed by the college.

How to get help

To ask questions about Python or homework, please use [our class Canvas discussion board](#) if your questions can be asked without revealing your homework solutions to your classmates. GSIs will check the discussion board approximately once a day. Alternatively, you can get answers to your questions about Python or homework during GSI office hours. Please do not use email to ask questions about Python or homework problems.

To ask questions about the lecture (or the lecture slides, or the textbook), please visit me during my office hours. For quick questions, I am also generally happy to talk immediately after class. Please do not use email to ask questions about course material.

For questions/concerns about grading, please first contact our GSI, either by attending GSI office hours or by email. If you wish to request a re-grading of your work, you must do so within one week of when the original grade was issued. If you are not satisfied with the answer you get from a GSI and wish to contest the GSI's grading, please [email me](#) or visit me during office hours. I will generally defer to our GSI's judgment on subjective matters (e.g., the amount of partial credit).

For personal questions (e.g., concerns about keeping up with the class, extended illness), please visit me during my office hours or [email me](#) to schedule an appointment.